

Prof. Dr. Orestis Terzidis



ENTREPRENEUR, ENTREPRENEURSHIP



Important Sources

„The entrepreneur **shifts economic resources** out of an area of lower and into an area of **higher productivity**.“

Jean Baptiste Say: Traité d'économie politique, 1803



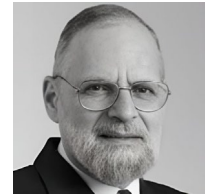
Entrepreneurship is about **new factor combinations** leading to new **products**, production **methods** or new **markets**. It is about **creative destruction**.

Joseph Schumpeter: Theorie der wirtschaftlichen Entwicklung 1911



Entrepreneurs are **alert discoverers** who, by **exploiting market opportunities** and **correcting price discrepancies**, incrementally guide **markets toward equilibrium**.

Israel Kirzner, according to Shane, & Venkataraman (2000)



Entrepreneurs...

...are persons who seek to establish an **organized activity** and **create value** for stakeholders by finding **new resource arrangements** and by introducing **new products and processes** or identifying **new markets** to serve the **needs of a target group**.

... **devote time and effort** and **assume** the necessary financial, psychic, and social **risks**; in case of success they receive the monetary and personal **satisfaction**.

(Terzidis, 2017)

(drawing from Ahmad, Hoffmann 2008, p. 8), (Hisrich, Ramadani 2017, pp. 4–5) and (Byers et al. 2011, Pos. 2905)

Entrepreneurship...

...is the phenomenon associated with the **enterprising human action** in pursuit of the generation of value.

(Terzidis, 2017), (Hisrich, Ramadani 2017, pp. 4–5) and (Byers et al. 2011, Pos. 2905)

Entrepreneurship is like giving birth to a child.
The mother's love overcomes the risk of life.
The entrepreneur's passion overcomes the risk of loss.

www.praveenben.com (October 2015)



Key Activities

“Entrepreneurs identify opportunities, mobilize resources, execute on their vision and manage risks.”

(Byers et al. 2011, Pos. 497)

“Entrepreneurship is focused on the creation of a new enterprise that serves society and makes a positive change.”

(Byers et al. 2011, Pos. 506)



Successfull entrepreneurs

Amazon

- Internet Retail, founded in 1994
- Revenue: US\$ 280.522 billion (2019)
- Number of employees: 1,000,000 (July 2020)
- One of the most influential economic and cultural forces in our times
- Estimated personal wealth: >200 Billion US-Dollars.

Jeff Bezos (*12.01.1964)



Source: Amazon | Image Source: forbes.com

Successfull entrepreneurs

Grameen Bank

- Microcredit and microfinance, founded in 1983
- 2,568 Branches in Bangladesh
- Number of employees: 20,138 (2018)
- Nobel Peace Prize 2006

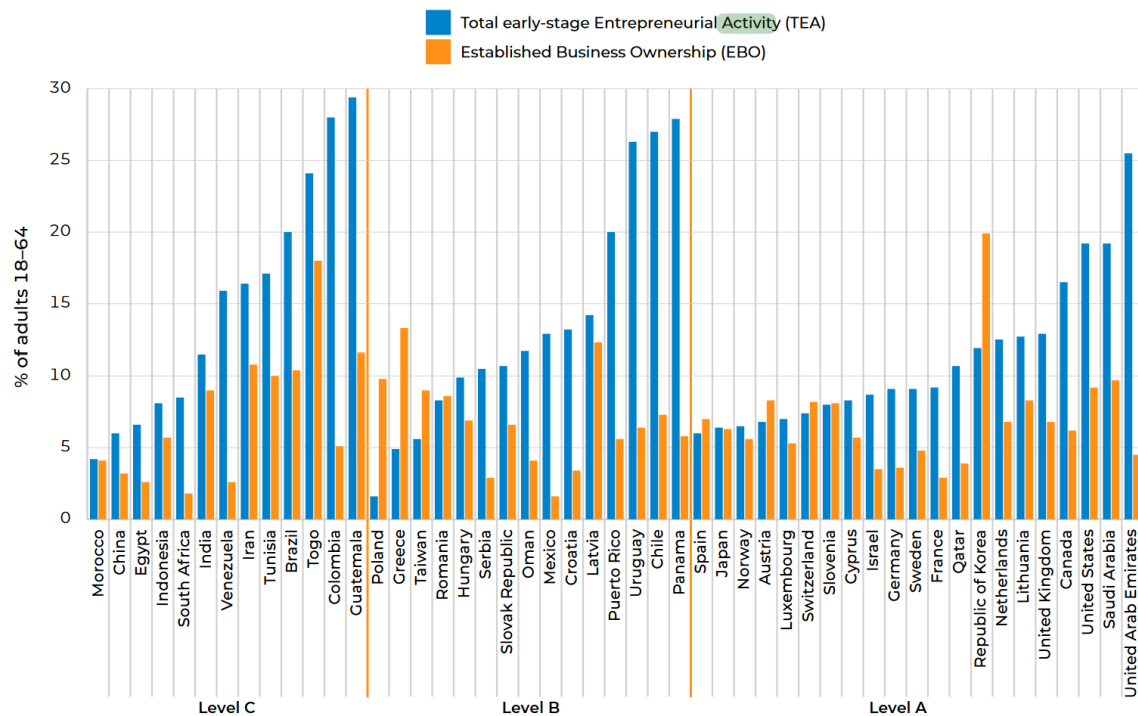
"for economic & social development from below"

Muhammad Yunus (*28.06.1940)



Source: Grameen Bank | Image Source: yunus.org

Global Entrepreneurship Monitor 2023



■ Total early-stage Entrepreneurial Activity (TEA) rate (% adults)

DSM 2023: Degrees of Founders

Management & Economics

Engineering

Computer Science &

Natural Sciences

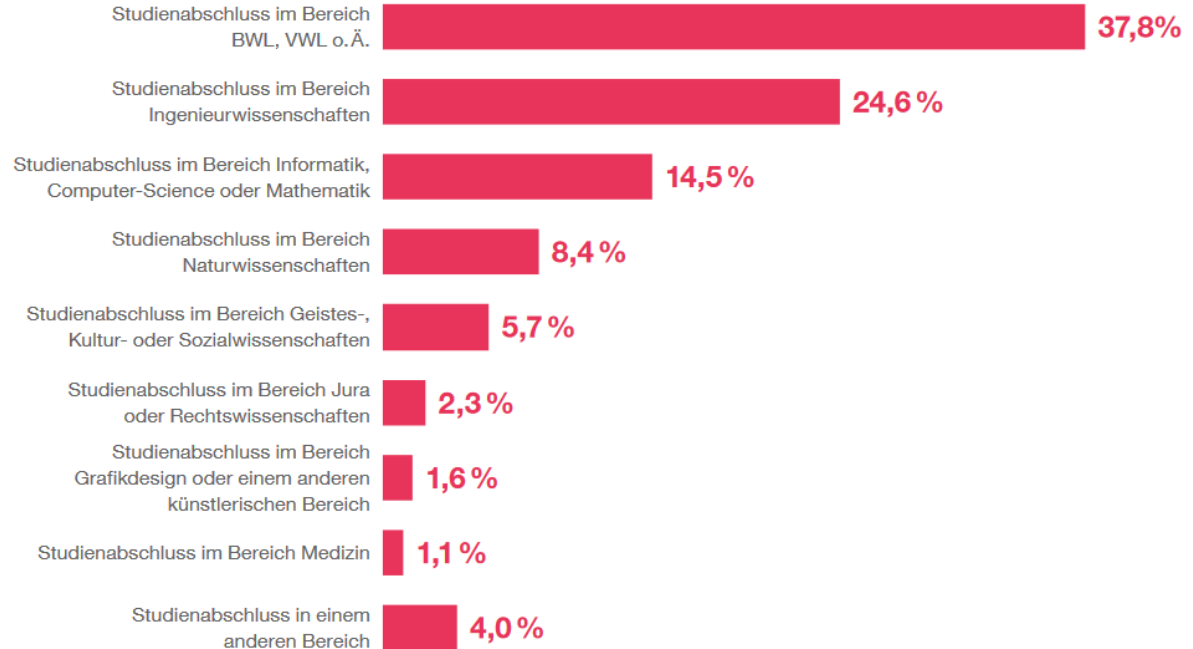
Humanities

Law

Design

Medicine

Other



Empirical Evidence: Traits play some Role

- Some traits significantly correlated with entrepreneurial behavior (business creation, business success).
- These relationships were of moderate size.

Rauch & Frese (2007)





Traits mentioned in the literature

Personality Traits	Authors
Conscientiousness	Zhao & Seibert, 2006, p.259
Openness (to experience)	Zhao & Seibert, 2006, p.259
Need for achievement	Rauch & Frese, 2007, p.353, Begley & Boyd, 1987, p.88, Baum, Frese & Baron, 2014, p.12-19
Generalized self-efficacy	Rauch & Frese, 2007, p.353, Baum, Frese & Baron, 2014, p.12-19
Innovativeness	Rauch & Frese, 2007, p.353, Baum, Frese & Baron, 2014, p.12-19
Risk-taking (propensity)	Begley & Boyd, 1987, p.88, Baum, Frese & Baron, 2014, p.12-19
Need for autonomy	Rauch & Frese, 2007, p.353, Baum, Frese & Baron, 2014, p.12-19
Proactive personality	Rauch & Frese, 2007, p.353
Stress-tolerance	Rauch & Frese, 2007, p.353
Locus of control	Baum, Frese & Baron, 2014, p.12-19
Tolerance of ambiguity	Begley & Boyd, 1987, p.88

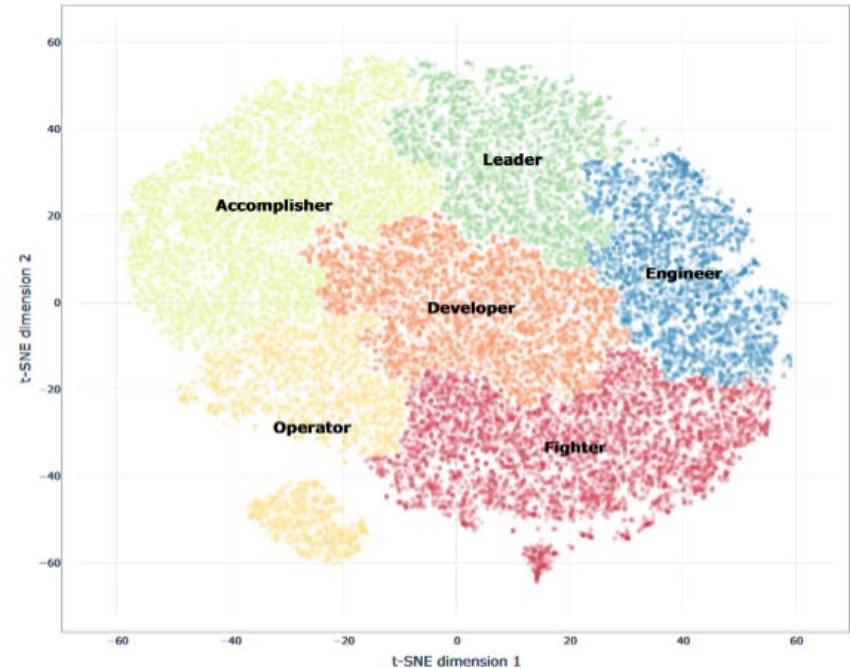


Founder personalities: Data Sources and Analysis

Two datasets:

- Twitter user profiles to infer individual personality facets
- Crunchbase company profiles on founding teams and company success
- 21,187 startup companies used in the success prediction model
- Correlated personality traits and startup success

McCarthy et al. (2023), „The Impact of Founder Personalities on Startup Success“.





Founder personalities are a key determinant

Fighter, Operator, Accomplisher, Leader, Expert/Engineer, Developer

Founder type		Distinctive Personality Traits
Clustered by personality (Cluster code)		Personality traits of founders in this cluster (Big Five facets)
Fighter	(#5)	Emotional stability (anger, anxiety, depression, immoderation, self-consciousness, vulnerability)
Operator	(#4)	Highest in conscientiousness in the facet of orderliness and high agreeableness in the facet of humility for founders in this cluster.
Accomplisher	(#0)	Highly extraverted (all facets) and conscientious (five facets)
Leader	(#2)	Highest in openness in the facets of artistic interests and emotionality also highest in agreeableness in facets of altruism and sympathy.
Expert/Engineer	(#1)	Highest in openness in the facets of imagination and intellect.
Developer	(#3)	'Middle child' cluster—no facets are maximums or minimums, but it shares characteristics similar to fighters but higher in extraversion.

McCarthy et al. (2023), „The Impact of Founder Personalities on Startup Success“.

Career Reasons of Nascent Entrepreneurs

Nascent entrepreneurs have similar reasons for career choice as working age adults.

There is no significant difference with respect to self-realization, financial success, innovation, and independence. However, nascent entrepreneurs rated significantly lower on roles and recognition.

Carter et al. (2003)

- **Independence** (freedom & control in the use of one's time)
- **Self-realization** (pursue self-directed goals)
- **Financial success** (earn money, achieve financial security)
- **Recognition** (have status & approval by others)
- **Role** (follow family traditions or follow example of others)
- **Innovation** (accomplish something new)

SOCIAL NETWORKS

“Jack-of-all-Trades” (JAT) Theory (Lazear, 2002)

Entrepreneurs as Generalists:

- Handle varied tasks: finance, marketing, operations, management
- Deep expertise in a single area is insufficient

Weakest Link Principle:

- Success limited by weakest skill
- Entrepreneurs must ensure all essential skills meet a critical threshold

Human Capital Investment Patterns:

- Entrepreneurs invest in balanced skill sets during education and career
- Employees/specialists focus on single area investments



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Social Capital Thoery

Coleman (1988):

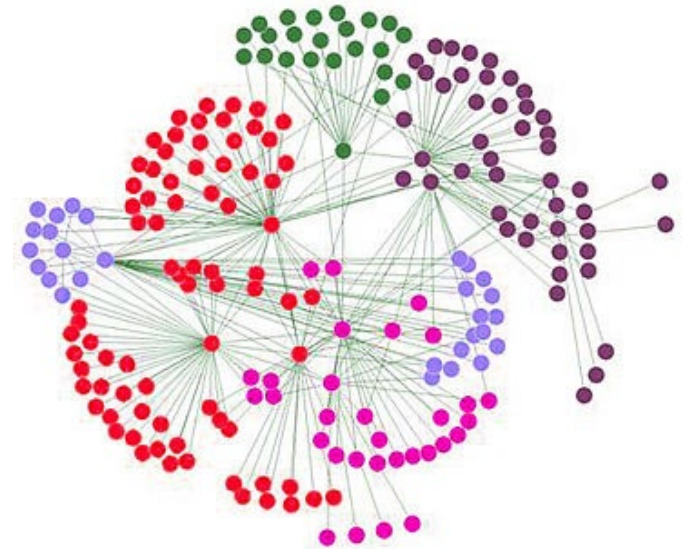
- Connections with others form "social networks"
- The structure and embeddedness of these networks are valuable resources

SN Definition (Hoang & Antoncic, 2003):

- A set of actors (individuals or organizations)
- A set of linkages between the actors

Importance of SN (Brass et al., 2004)

- Is highly relevant for foundation and growth of new firms

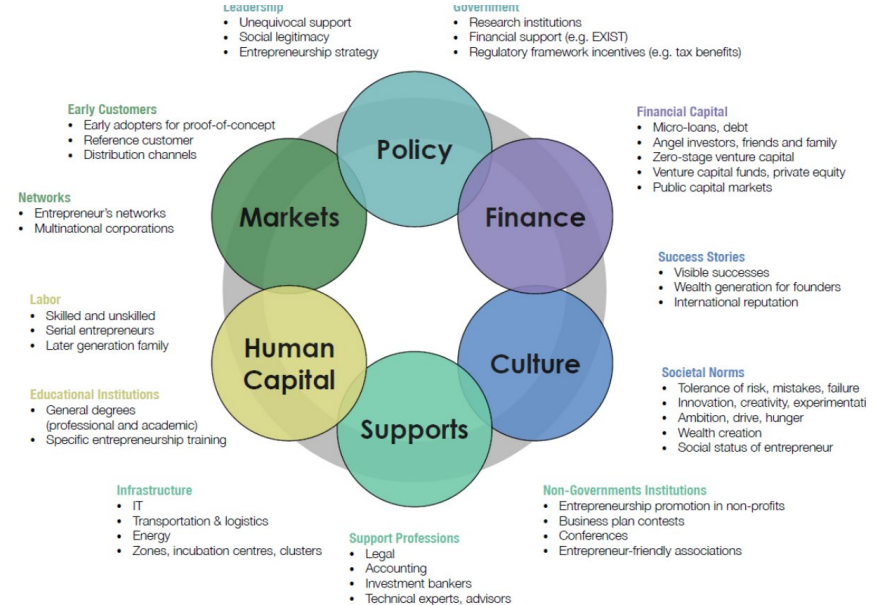


Entrepreneurial Ecosystems

An entrepreneurial ecosystem is a community composed of various independent factors that interact within a geographical area to foster the creation and growth of new businesses.

It contains policy, finance, markets, Human Capital, Culture, and Supports.

(Isenberg, 2011)



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ROLE IN THE ECONOMY



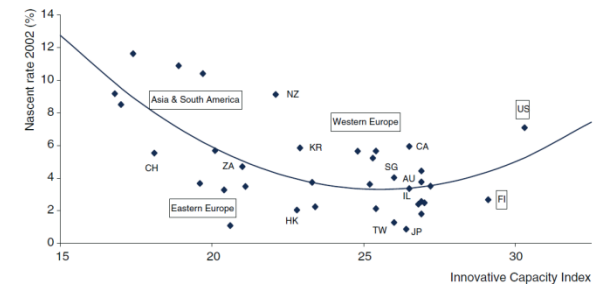
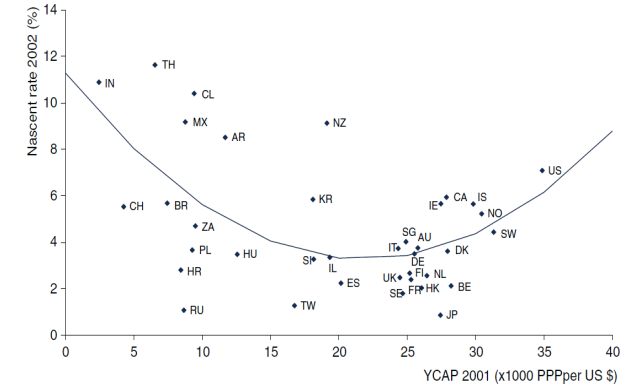
Entrepreneurship is considered key to economic performance, in particular with respect to innovative change, playing an important structural and dynamic role in all economies.

Fostering Entrepreneurship and Firm Creation as a Driver of Growth in a Global Economy, p. 5



Entrepreneurship and Growth/Innovation

- Wennekers, van Wennekers, Thurik, & Reynolds (2005) investigated the relationship between nascent entrepreneurship rates and, per capita income and innovative capacity in 36 countries.
- They found a U-shaped relationship for both correlations.
- The authors suggest that the positive effects of entrepreneurial activities are stronger in developed countries than in developing countries.



GDP Growth is Correlated with Growth Potential TEA

TABLE I
Result of linear regression

df = 32	Type of TEA measure		Opportunity TEA		Necessity TEA		Overall TEA	
	High Growth Potential TEA							
Average (%) across 37 countries	1.03		5.62		1.95		7.99	
	Value	Sig.	Value	Sig.	Value	Sig.	Value	Sig.
Adj R sq (control)	0.559		0.559		0.559		0.559	
Adj R sq	0.648		0.603		0.605		0.602	
F change	5.305**	0.010	2.893*	0.070	2.971*	0.066	2.858*	0.072
F	17.571**	0.000	14.676**	0.000	14.769**	0.000	14.634**	0.000
	Std. Beta	Sig.	Std. Beta	Sig.	Std. Beta	Sig.	Std. Beta	Sig.
Control variables								
(Constant)	t = 5.995**	0.000	t = 4.048**	0.000	t = 4.019**	0.000	t = 3.593**	0.001
Log of base year GDP per worker	-0.536**	0.000	-0.475**	0.001	-0.529**	0.001	-0.491**	0.001
Growth in capital investment per worker 1997/1998-2001/2002	0.602**	0.000	0.625**	0.000	0.597**	0.000	0.614**	0.000
Predictors								
Entrepreneurship (TEA)	0.207*	0.050	0.029	0.806	-0.066	0.663	-0.007	0.958
Innovation (Ratio of USPTO granted Patents to GDP 1997-2001)	0.284**	0.013	0.275**	0.023	0.268**	0.028	0.275**	0.023

Dependent variable = growth in GDP per worker 1997/1998-2001/2002.

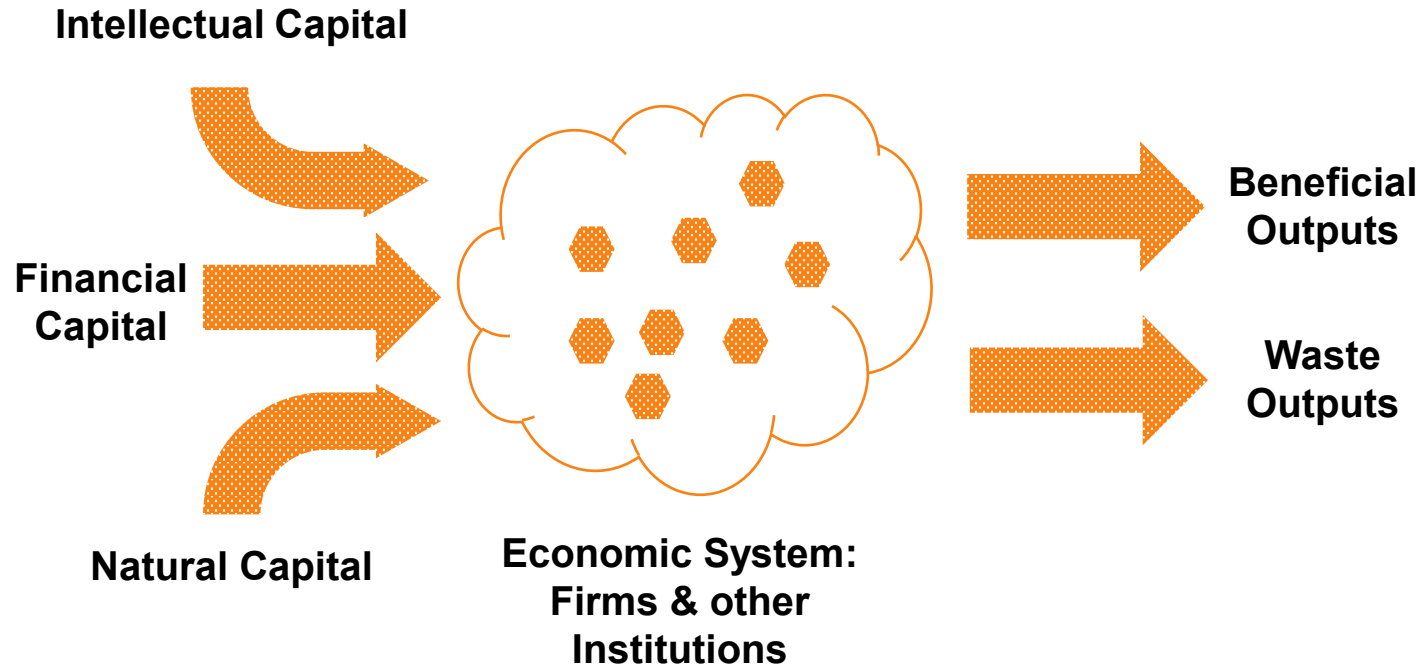
* Significant at 10%.

** Significant at 5%.

- Wong, Ho, & Autio (2005)
- Study based on GEM data from 37 countries in 2002
- Four types of TEA: high growth potential, necessity, opportunity and overall TEA.
- TEA = total entrepreneurial activity



A simple model of the economy



Forms of Capital

Natural Capital

- Raw materials
- Energy (wind, water, sun, waves)
- Biological yield
- Pollution absorption capacity
- Land

Financial Capital

- Money
- Bonds & Securities

Intellectual Capital

- Human Capital: Knowledge, Skills, Attitudes

Organizational Capital

- Hardware & Machines
- Software & Databases
- Patents
- Processes & methods

Social Capital

- Quality of stakeholder relationships



Three reasons for studying Entrepreneurship

Entrepreneurship is a mechanism by which society **converts technical information into products and services**.

Entrepreneurship is a mechanism through which **temporal and spatial inefficiencies** in an economy are discovered and mitigated.

Entrepreneurially driven innovation in products and processes is the crucial **engine driving the change process in the economy**.

Shane & Venkataraman (2000)

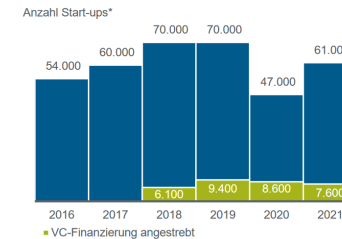
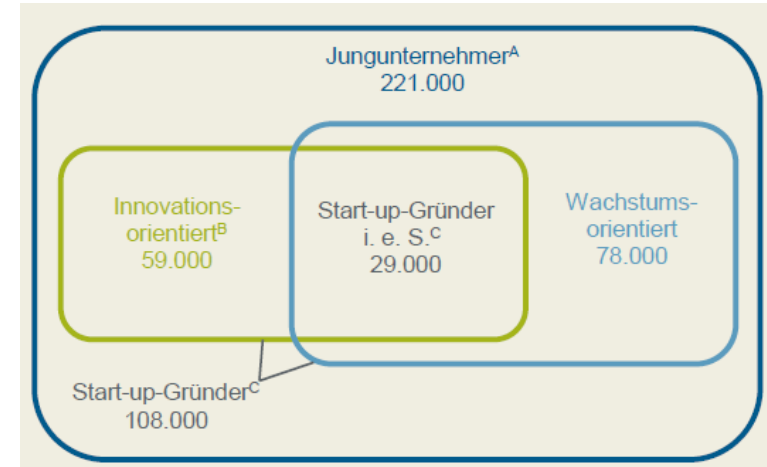


Entrepreneurship in Germany

In 2017, there were around 108,000 start-up founders with 60,000 young companies in Germany.

Start-ups are innovative or growth-oriented young companies.

KfW-Start-up-Report





Creative Destruction (Schöpferische Zerstörung)

Static paradigm

- World of routine transactions
- Efficiency and price are main drivers of competition
- Gradual changes, only few options are explored

Dynamic paradigm

- New innovative firms forming and growing
- Old established business models declining and failing
- Disequilibrium & disruption



A Startup is an act of rebellion (Ken Singer)



- A startup proposes to do something new.
- It is a rejection—the current way is insufficient.
- But some people/companies are invested in the current system and they will resist.
- Your success could mean their demise.

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ENTREPRENEURIAL OPPORTUNITY



„Die Gelegenheit beim Schopfe packen“

Kairos (καιρός) is an ancient Greek word meaning the **right or opportune moment**.

Kairos signifies a time in between, a **moment of indeterminate time** in which **something special happens**.

Kairos (καιρός) also means weather.

Fresco from Francesco Salviati, Rome, 1552/54.



Definition – Business Opportunity

An opportunity is a **timely and favorable juncture of circumstances** providing a **good chance** for a **successful venture**.

Objectivism vs. Constructivism

Discovery Theory (Shane, 2000; Eckhardt & Shane, 2010):

*[...] opportunities are **like lost luggage in a train station**; they exist, just waiting to be claimed by alert individuals who know of their existence.
(Alvarez et al., 2010)*

Creation Theory (Alvarez & Barney, 2007; Alvarez et al., 2010;
Alvarez et al., 2013):

*The entrepreneur's goal is **to construct, deconstruct, and reconstruct an existing reality** so as to form a new reality and thus opportunity.*

Nine Categories of Opportunity

- Increasing the value of a product or service
- New applications of existing means or technologies
- Creating mass markets
- Customization for individuals
- Increasing reach
- Managing the supply chain
- Convergence of industries
- Process innovation
- Increasing the scale of the firm

| Source: Byers et al., Section 2.1



Characteristics of attractive Opportunities

- **Timely** – a current need
- **Solvable** – can be solved in the near future with accessible resources
- **Important** – on the priority list of the customer
- **Profitable** – adequate willingness to pay
- **Context** – favorable regulatory and industry situation

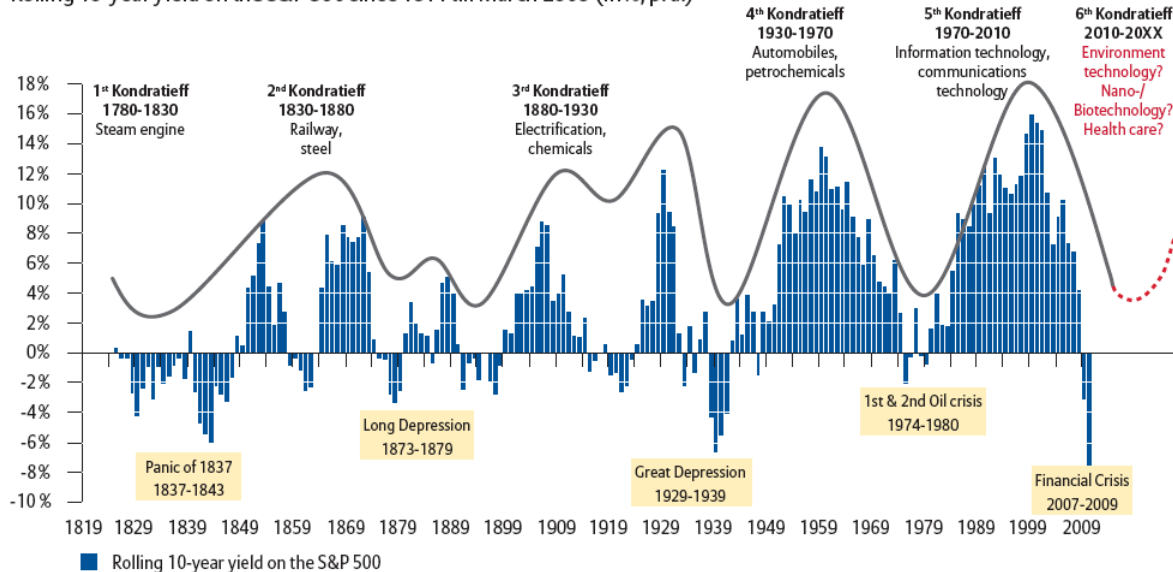




Kondratieff Cycles

Kondratieff cycles – long waves of prosperity.

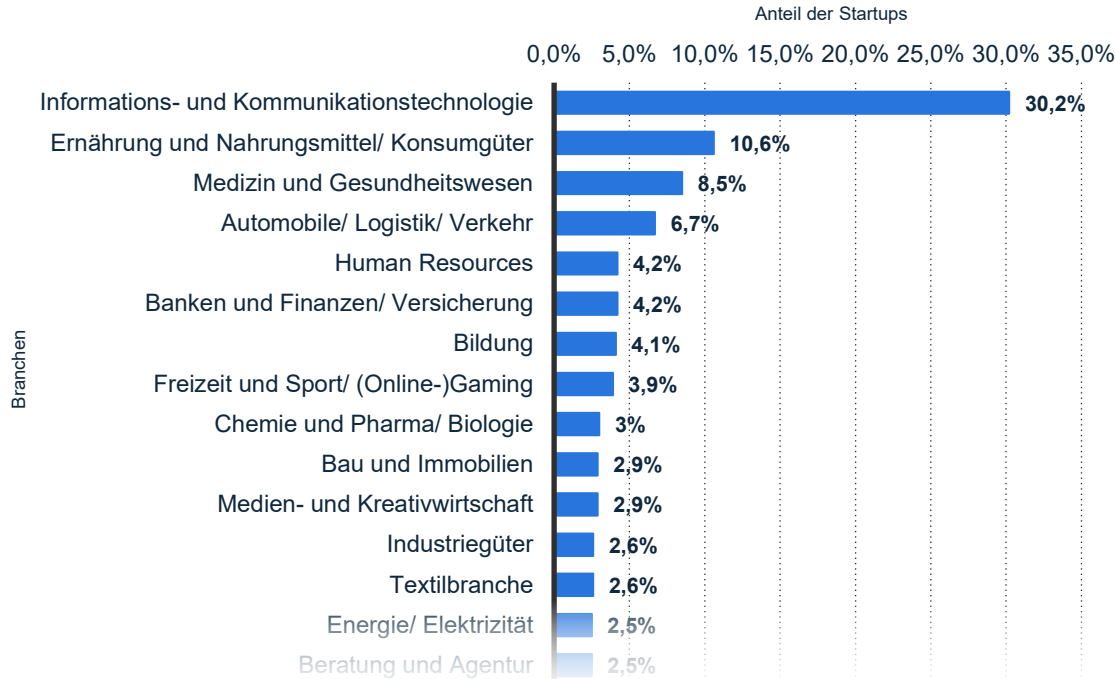
Rolling 10-year yield on the S&P 500 since 1814 till March 2009 (in %, p. a.)



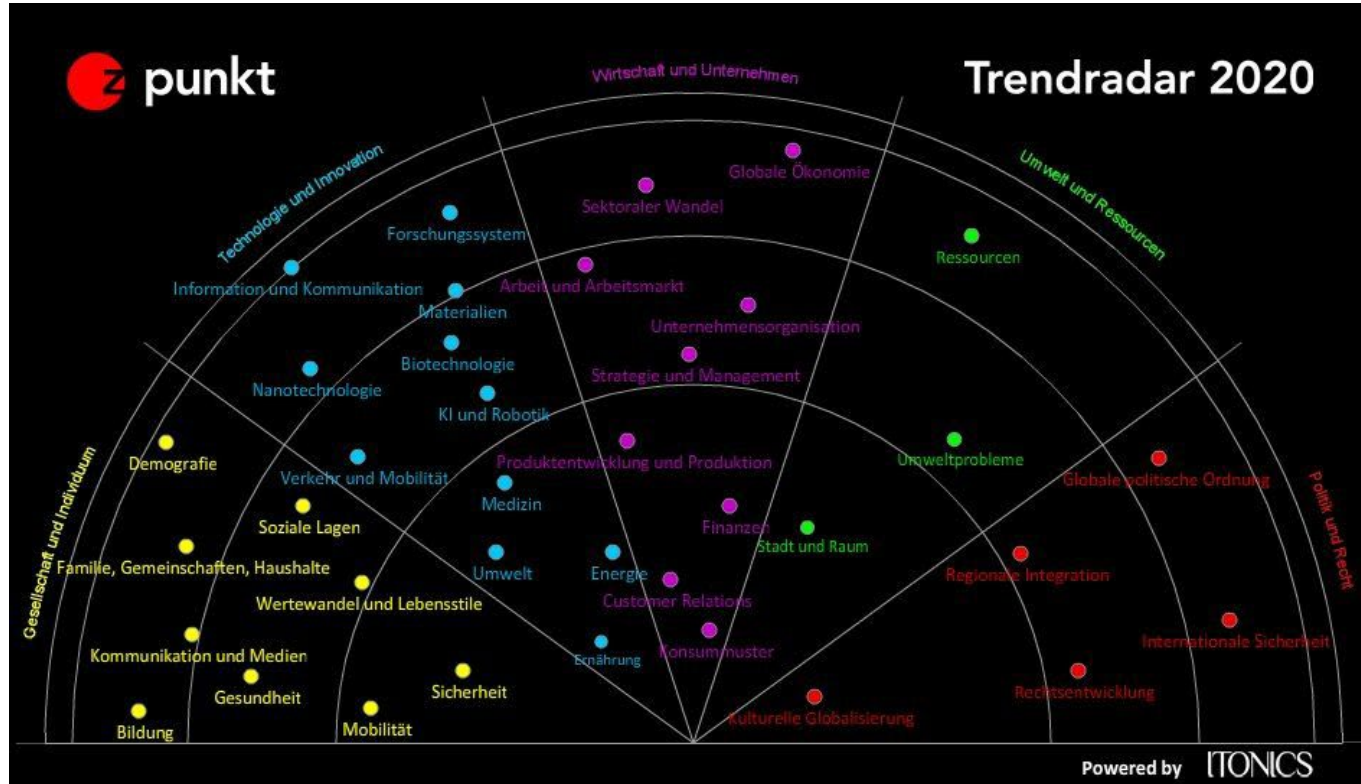
Source: Datastream; Illustration: Allianz Global Investors Capital Market Analysis

- **Steam engine**
- **Railway, Steel**
- **Electrification, chemicals**
- **Automobiles, petrochemicals**
- **ICT**
- **Clean Tech, Nano, Bio, Healthcare**

DSM* 2019: Industry Distribution



- **ICT**
- **Nutrition**
- **Medical & Health**
- **Mobility**
- **Human Resources**
- **Financial Services**
- **Education**
- ...





SUSTAINABLE DEVELOPMENT GOALS



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ENTREPRENEURSHIP EDUCATION

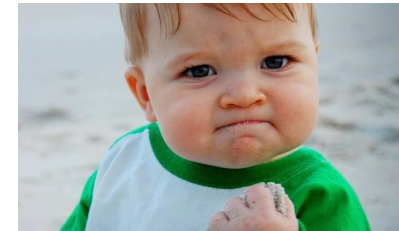
We define a competence as the disposition **to generate adequate actions** to **responsibly solve problems** in variable situations.

This ability is based on **knowledge, skills, and attitudes**.

Tittel & Terzidis, 2020

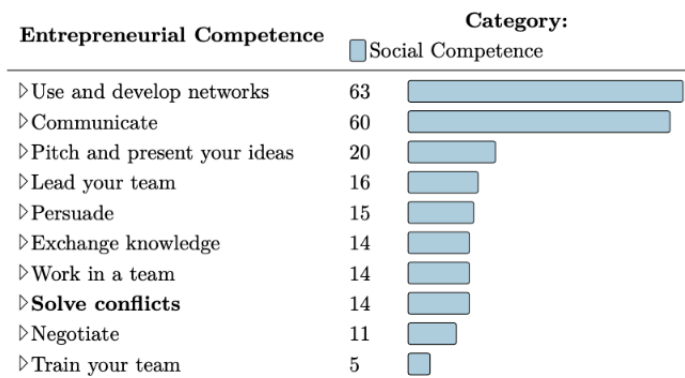
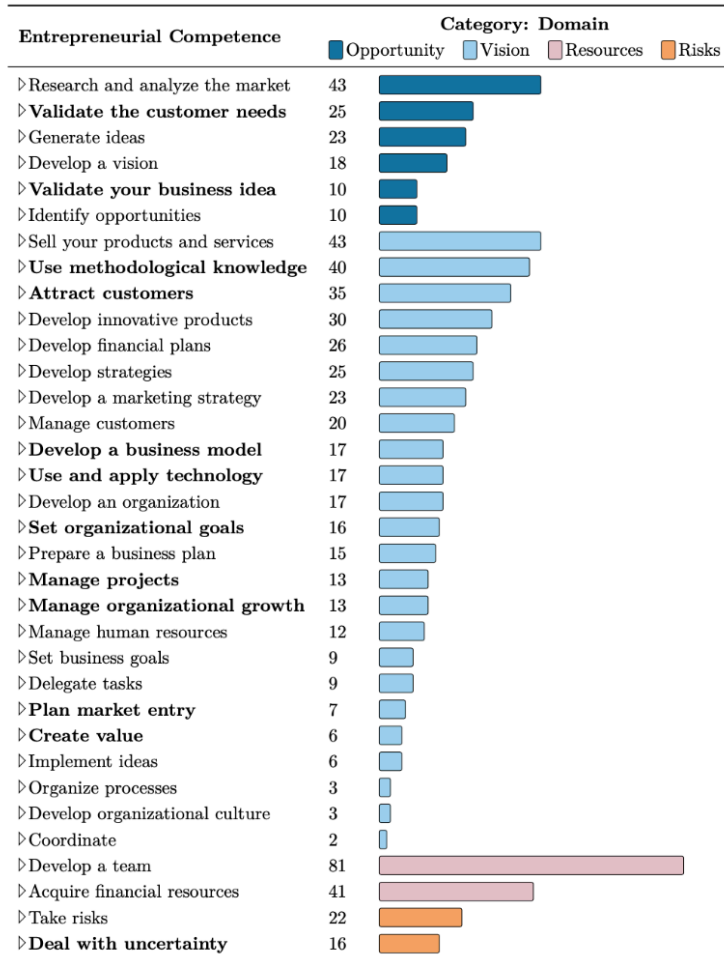
Competence

- **Knowledge** is the body of facts, principles, theories and practices that is related to a field (inspired by EQF).
- **Skill** is the ability to apply knowledge and use know-how to complete tasks (inspired by EQF).
- **Attitude** is a disposition to respond favorably or unfavorably to an object, person, institution, or event (Ajzen, I. 2005, p.3).



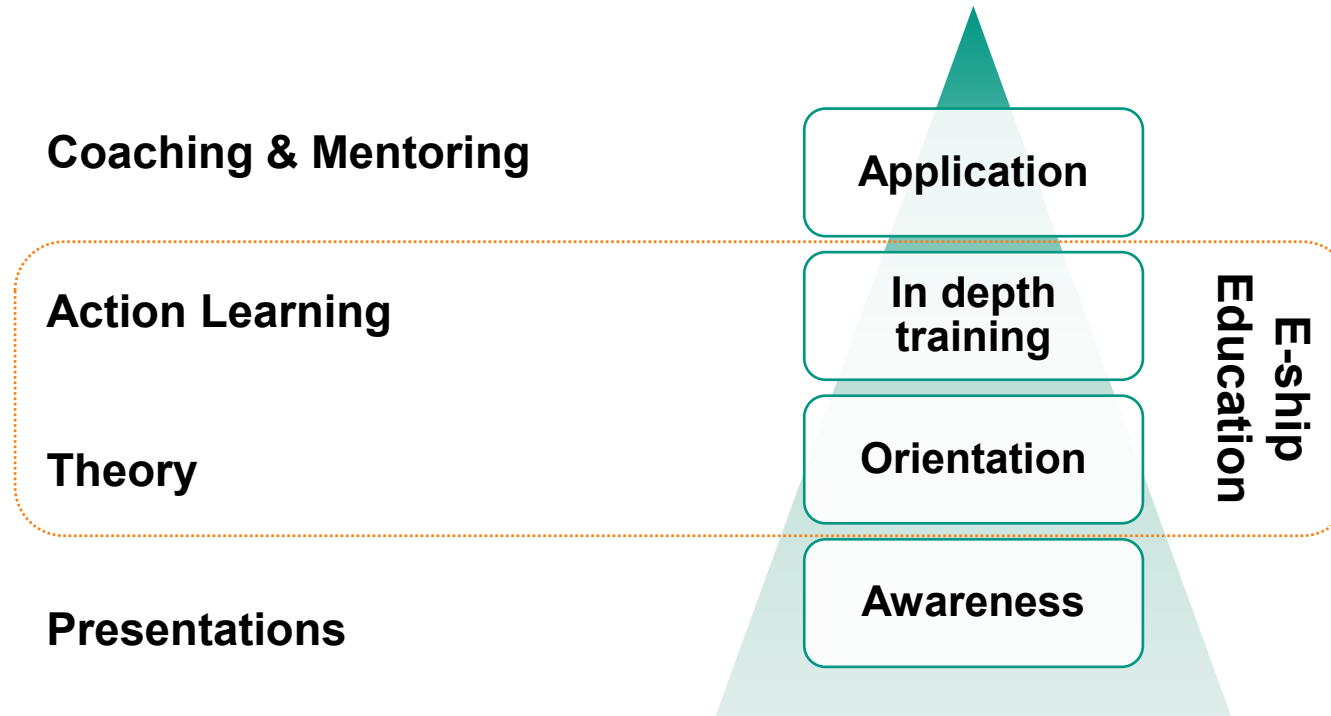
An **entrepreneurial competence** is the specific set of **domain**, **relational**, and **personal** competences needed **to generate entrepreneurial action**.

Tittel & Terzidis, 2020





From Awareness Building to Application



Action Learning

Students learn effectively at deeper levels if they are actively engaged in the process.

Stages :

- **Listen**: Impulse
- **Do**: Activity & experience
- **Reflect**: what, how, why, how else
- **Internalize**: Theorize & practice



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Thank You!

